MODULE 4

Fuzzy Logic

What is Fuzzy Logic?

- Fuzzy Logic is a multi-valued logic system that deals with approximate reasoning.
- Unlike classical (crisp) logic which uses True/False (1 or 0).
- Fuzzy logic allows degrees of truth between 0 and 1.
- Introduced by Lotfi A. Zadeh in 1965.
- Designed to handle uncertainty and imprecision in real-world problems.

Example of Crisp vs Fuzzy Logic

Crisp Logic:

- Temperature = Hot → True if > 35°C, otherwise False

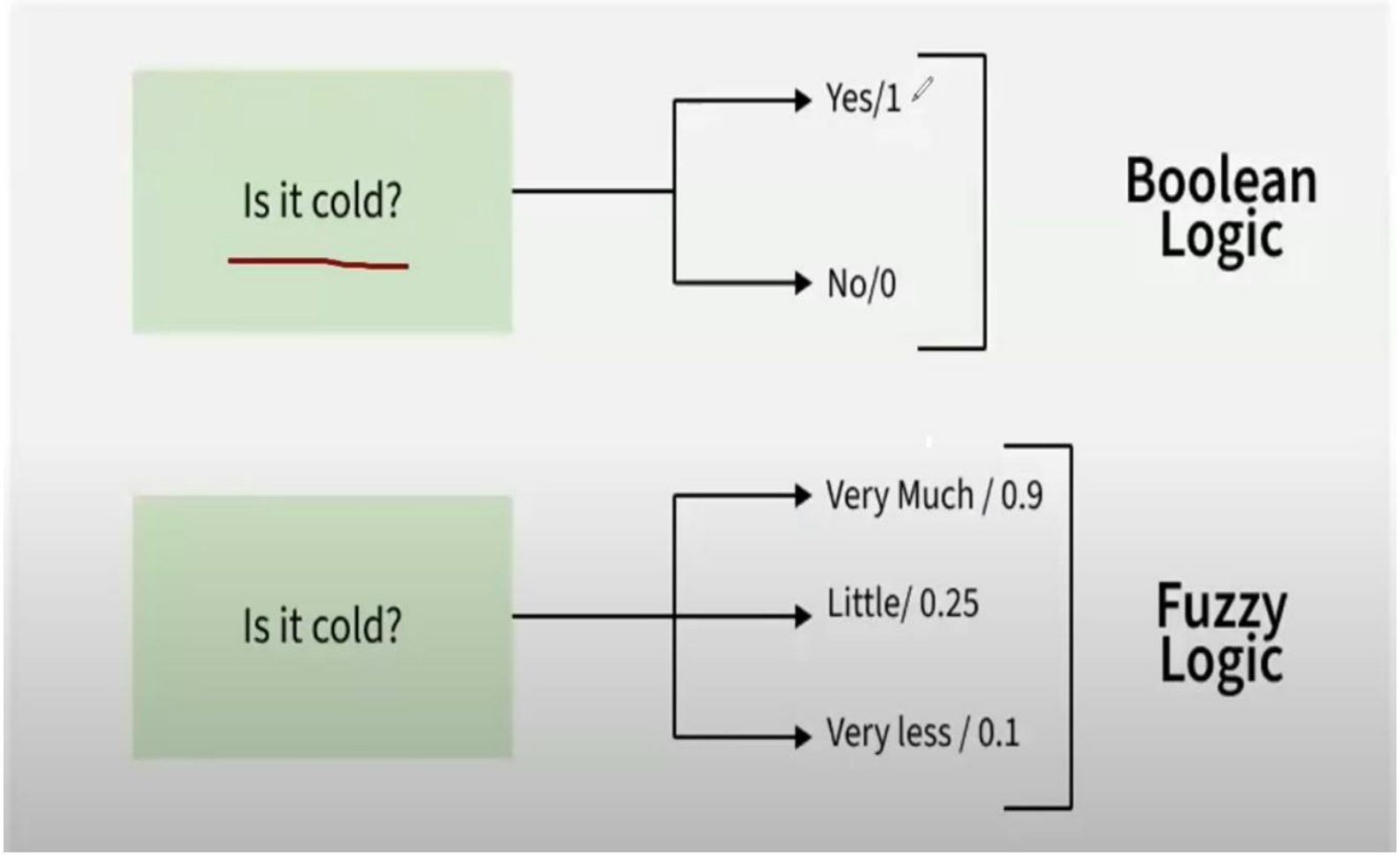
Fuzzy Logic:

- Temperature = Hot → 0.7 (partially hot) at 32°C
- Temperature = Hot → 1 (fully hot) at 38°C

Why Fuzzy Logic?

- Handles uncertainty and imprecision.
- Real-world situations are not always completely true or false.
- Allows values between 0 and 1 to represent partial truth.
- Helps systems make flexible decisions.
- Used in control systems, AI, medical diagnosis and image processing.

Boolean Logic vs Fuzzy Logic



Key Concepts of Fuzzy Logic

1. **Membership Function** – Determines the degree to which an input belongs to a set.
2. **Fuzzy Sets** – Elements have partial membership between 0 and 1.
3. **Linguistic Variables** – Variables described using words like cold, warm, hot.

Membership Function Example

- Example: Temperature in a fuzzy system
- 70°F may belong:
- • 0.8 to 'Warm'
- • 0.2 to 'Cool'
- This shows partial membership in multiple categories.

Fuzzy Sets Example

Example: Age classification

Age 30 → Membership 0.7 in 'Young'

Age 50 → Membership 0.3 in 'Young'

- This means the classification is not strictly fixed.

Linguistic Variables

- Variables described using words instead of exact numbers.

- Example:
- Temperature = {Cold, Warm, Hot}
- These terms help systems reason in a human-like way.